SIMATS ENGINEERING

Department of Information Technology

CSA5103– Cryptography and Network Security

ASSIGNMENT 1 – CO1

Design, Analyse and Validate Classical Encryption Techniques for Securing Sensitive Information

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for ensuring data security.

Student Name	Register Number	Department / Section	Academic Year / Semester	Faculty Name	Nature of Assignment	Duration	Total Marks	Date of Submission	Date of Evaluation
G. Yashmita	192511032	CSE	2026 -27	Dr.R.Dharani	Individual	1 Week	50	10.10.26	11.10.26

MARK SHEET (Rubric-based Evaluation)

Task	Task Title	Max	Marks Awarded	Remarks / Evidence Expected
1	Cipher Design and Selection	5		Select and describe three suitable classical ciphers (Caesar, monoalphabetic substitution, and Vigenère or Hill); justify their use and limitations for the given sensitive message.
2	Encryption and Decryption	5		Show correct key selection, encryption steps, ciphertext, decryption steps and recovery of the original plaintext.
3	Key Analysis and Mathematical Basis	5		Explain key space, modular arithmetic or alphabet mapping used by each selected cipher; calculations must be traceable.
4	Security and Cryptanalysis	5		Analyse vulnerabilities such as brute-force search, frequency analysis, repeated-key patterns and known-plaintext attacks where relevant.
5	Comparison of Cipher Strength	4		Compare the selected methods using key space, implementation effort, likely attacks and suitability for sensitive information.
6	Algorithm / Pseudocode Formulation	4		Provide clear encryption and decryption pseudocode or flowcharts, including input validation and treatment of spaces/punctuation.
7	Worked Example and Verification	4		Use a sample message; verify that decrypt(encrypt(plaintext)) reproduces the original message under the stated rules.
8	Implementation and Testing	4		Implement at least two ciphers in a programming language; include source code, test cases, expected output and actual output.
9	Experimental Evaluation	3		Test multiple messages and key choices; present organised tables of ciphertext, recovered plaintext, execution time or operation counts.
10	Theory-to-Result Discussion	3		Explain how observed results match the cipher design and why classical ciphers are not suitable for modern high-security use.
11	Recommend and Defend a Design	4		Recommend a method for the assignment scenario and defend the decision using correctness, usability and security limitations.
12A	Re-evaluate: Impact Analysis	2		Discuss the impact of message length, key choice and repeated patterns on confidentiality.
12B	Re-evaluate: Attack Evaluation	2		Evaluate likely attacks against the selected ciphers and identify information an attacker could exploit.
12C	Re-evaluate: Adaptation or Redesign	2		Propose a safer modern replacement or layered approach, explaining why classical ciphers are only educational examples.
12D	Re-evaluate: Evidence-Based Defence	2		Defend conclusions using worked examples, test results, key-space reasoning and a comparison table.
	TOTAL	50		Marks to be awarded by the faculty.

OVERALL REMARKS

Faculty feedback:

Student Signature	Faculty Signature	Program Director
		

MANDATORY SUBMISSION EVIDENCE CHECK

✓ = present P = partly present ✗ = missing. Mark the status after checking the student's submitted report and supporting evidence.

Task	Evidence Expected	Status	Note
1	Cipher selection, design rationale and assumptions	☐	Three ciphers described and justified
2	Encryption and decryption working shown	☐	Steps, key and plaintext/ciphertext included
3	Mathematical basis and key-space explanation	☐	Calculations are traceable
4	Security analysis and likely attacks	☐	Relevant weaknesses identified
5	Cipher comparison table	☐	Criteria applied consistently
6	Pseudocode or flowchart	☐	Encryption and decryption steps included
7	Worked example and correctness verification	☐	Recovered plaintext matches original
8	Source code and reproducible tests	☐	Program, test inputs and outputs included
9	Experimental results in tables	☐	Multiple inputs or keys evaluated
10	Theory and experimental results compared	☐	Observations interpreted
11	Recommendation supported by analysis	☐	Trade-offs and limitations stated
12A–D	Impact, attack evaluation, redesign and evidence-based defence	☐	All four parts addressed

IMPLEMENTATION EVIDENCE CHAIN (Tasks 8–9)

Evidence Item	Status	Observation
Source code	☐	Readable source code for each implemented cipher; language and execution instructions stated
Test inputs	☐	Plaintexts, keys and edge cases recorded
Expected behaviour	☐	Expected ciphertext and recovered plaintext documented
Actual output	☐	Program output or screenshots included
Intermediate evidence	☐	Key mapping, modular steps or intermediate transformations shown
Boundary / failure tests	☐	Empty input, spaces, punctuation, invalid keys and repeated characters considered
Interpretation	☐	Results connected to cipher properties and security limitations
Code, data and report consistent	☐	Tables and claims match the submitted program output

SUBMISSION CHECKLIST (not separately marked)

Requirement	Status
All assignment tasks including 12A–12D answered	☐ Yes ☐ Partly ☐ No
Encryption and decryption shown for each selected cipher	☐ Yes ☐ Partly ☐ No
Keys and assumptions clearly stated	☐ Yes ☐ Partly ☐ No
At least two ciphers implemented and tested	☐ Yes ☐ Partly ☐ No
Same plaintext conventions used consistently	☐ Yes ☐ Partly ☐ No
Source code and reproducible test procedure provided	☐ Yes ☐ Partly ☐ No
Tables / figures / program outputs included	☐ Yes ☐ Partly ☐ No
Cross-references and numbering correct	☐ Yes ☐ Partly ☐ No
No unfilled template placeholders in the student report	☐ Yes ☐ Partly ☐ No
Viva recommended where implementation or calculations need clarification	☐ Yes ☐ No

SIMATS ENGINEERING

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ASSIGNMENT 1

QUESTIONS

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security.

Assignment Title: Evaluate, Design and Defend a Secure Patient-Record Exchange Strategy for a Multi-Hospital Healthcare Network (Total Marks: 50)

Scenario — Confidential Patient Record Exchange

A regional healthcare network links one central hospital with 25 satellite clinics. Clinics send patient records, diagnostic reports, prescriptions and billing data to the central hospital over a public network and receive specialist opinions in return. Personnel hold different clearance levels: Medical Director (Top Secret), Specialist Doctors (Secret), Nurses and Lab Staff (Confidential), Billing and Front-Desk Staff (Restricted), Patients (Public).

The system must:

- Keep diagnoses, especially for sensitive conditions, confidential.
- Detect tampering with prescriptions, dosage and lab values.
- Prevent staff from reading or writing records beyond their clearance.
- Work in rural clinics where staff have limited technical training and hardware is basic.

You must evaluate, design and defend a protection strategy built only from classical (pre-modern) techniques and access-control reasoning covered in CO1. You are expected to judge how strong each technique actually is, not just describe it.

Design Challenge: An insider at a satellite clinic has been copying ciphertexts for months. Hundreds of messages under the same key are now exposed, and many begin with predictable text (e.g., "PATIENT ID", "DIAGNOSIS:", "PRESCRIPTION:"). The network also wants to embed a short report inside medical scan images sent between hospitals. Re-evaluate your design.

- Encrypt clinical notes (200 to 10,000 characters) using Caesar, monoalphabetic, Vigenère and columnar transposition, and derive each key space. Break at least three schemes by frequency analysis, Kasiski or index of coincidence, and rank the schemes by resistance.

- Extend the alphabet to digits and symbols (mod 36) and encrypt IDs and dosages such as PT-20481, AMOX 500MG using a Hill cipher on 3-character blocks. Prove the key's invertibility, compute one inverse, and recover the key by a known-plaintext attack.

- Apply columnar transposition to a lab table of 20+ rows and test whether a changed cell or a swapped row is detected. Design a classical checksum and show where an attacker defeats it.

- Encrypt a 256×256 or larger medical image with two classical methods (intensity shift mod 256 and pixel permutation) and verify decryption. Compare histograms and entropy, and explain which method leaks the image's structure.

- Hide a short report in an image by LSB embedding and in a text circular by a text method, then extract it. Measure PSNR, capacity and chi-square detectability, and test survival under JPEG compression.

- Decide 12+ access requests using Bell-LaPadula with compartments, including declassification and break-glass cases. Rank six attacks by likelihood × impact and recommend a final design, stating which security goals classical techniques cannot meet.

TASK 1 – Evaluate Threats, Attacks and Risks **4 Marks**

- Identify at least six security attacks relevant to the healthcare scenario.
- Classify each as passive or active (eavesdropping, traffic analysis, modification of prescriptions, masquerade as a doctor, replay of old orders, denial of service on the records system).
- Distinguish clearly between threat, vulnerability, attack and risk, with examples from this scenario.
- Rank the attacks by risk using a justified likelihood × impact rating. Consider patient-safety impact, not just privacy.
- Defend your ranking: which attack would you address first, and why?

Evidence Expected: A risk table with reasoned ratings and a justification of the priority order.

TASK 2 – Map and Judge Security Goals **3 Marks**

For each system function (record upload, prescription transmission, specialist consultation), decide which goals matter most: confidentiality integrity availability authentication non-repudiation

- Judge which goals classical ciphers can and cannot deliver.
- Justify why a correctly decrypted message does not prove it came from the right doctor, or that its dosage was not altered.

Evidence Expected: A goal-to-function analysis with an evaluation of the gaps classical techniques leave.

TASK 3 – Apply and Critique the Bell-LaPadula Model **5 Marks**

- Define the security levels and subject/object assignments for the network.
- Apply the simple security property (no read up) and the *-property (no write down) to at least five
- access requests, including two that must be denied. Show the reasoning for each.
- Evaluate the model critically in a medical setting: what happens in an emergency when a nurse needs a Secret-level allergy record? How does the model handle a doctor who must write a note to a lower-level billing file? Discuss tranquility and trusted subjects.
- Judge the model's limitations (no integrity guarantees, covert channels, rigidity in emergencies) and argue whether Bell-LaPadula alone suits healthcare.

Evidence Expected: Worked access decisions, a justified critique, and a clear verdict on the model's suitability.

TASK 4 – Design the Cryptographic Strategies **4 Marks**

Select at least three meaningfully different classical schemes from: Caesar, monoalphabetic substitution, Vigenère (polyalphabetic), Hill cipher, a transposition cipher. For each, provide:

- The encryption and decryption algorithm (pseudocode or formula) and the key definition.
- A worked example on a 20+ character message from the scenario.
- A justification for including it in your comparison.

Evidence Expected: Three distinct schemes with algorithms, worked examples and justified selection.

TASK 5 – Mathematical Analysis of the Schemes **5 Marks**

- Derive the key-space size of each scheme (e.g., 25, 26!, 26^m, number of invertible n×n matrices mod 26, n! permutations).
- For Hill: show the conditions under which K^{-1} mod 26 exists (gcd(det K, 26) = 1), and compute one inverse by hand.
- Express each encryption as a modular function (e.g., C = (aP + b) mod 26 for an affine variant, C = KP mod 26 for Hill).
- Judge whether a large key space guarantees strength. Support your answer with the monoalphabetic example.

Evidence Expected: Correct derivations with a reasoned judgment on key space versus real security.

TASK 6 – Evaluate Cryptanalytic Resistance **4 Marks**

Attack each chosen scheme using a technique suited to it:

- Brute force, frequency analysis, Kasiski examination / index of coincidence, known-plaintext attack on Hill, digram analysis.
- Explain why each attack succeeds or fails, in terms of letter-frequency preservation, key repetition and linearity.
- Rank the schemes by resistance and justify the ranking with evidence.

Evidence Expected: Worked attacks with reasoning, and a ranking of schemes supported by the results.

TASK 7 – Assess Transposition and Product Ciphers **3 Marks**

- Implement a columnar (or rail-fence) transposition and explain how it diffuses plaintext structure.
- Evaluate whether *substitution followed by transposition* is stronger than either alone. Justify with letter-frequency evidence and an attack attempt.

- Judge whether the added complexity is practical for rural clinic staff doing manual or low-resource encryption.

Evidence Expected: A reasoned evaluation of the combined scheme against its components.

TASK 8 – Evaluate Steganography **3 Marks**

Design a steganographic method (e.g., LSB embedding in a scan image, or text-based hiding) for carrying a short report inside a medical image.

- Compare it with encryption on capacity, imperceptibility, robustness and detectability.
- Evaluate the risk specific to medical imaging: could embedding alter pixel data enough to affect a diagnosis? Judge the trade-off.
- Judge when steganography is appropriate, and argue why hiding is not a substitute for encrypting. Evaluate an encrypt-then-embed approach.

Evidence Expected: A criteria-based comparison with a justified conclusion.

TASK 9 – Implement and Test **4 Marks**

Implement all chosen schemes, the cryptanalysis tools and (where feasible) the steganography method, in a language of your choice.

- Provide correctness tests: decrypt(encrypt(m)) = m, including edge cases (digits in patient IDs, punctuation, non-multiple block length for Hill).
- Use identical plaintexts of 100, 1,000, 10,000 and 100,000 characters across schemes.
- State the testing procedure clearly so it can be reproduced.

Evidence Expected: Working code with a documented, reproducible test procedure.

TASK 10 – Record and Interpret Experimental Evidence **3 Marks**

Present data in tables and graphs for:

- Letter-frequency histograms of plaintext versus each ciphertext.
- Index of coincidence per scheme.
- Encryption/decryption time against message size.

- Attack success rate against ciphertext length.

Evaluate what the data shows. Do not just present it.

Evidence Expected: Clear, comparable results with interpretation tied to Tasks 5 and 6.

TASK 11 – Recommend and Defend the Final Design **4 Marks**

Recommend one scheme, or a layered combination, together with the Bell-LaPadula policy.

- Justify against the alternatives using your mathematical and experimental evidence.
- State the residual risks and limitations honestly. Judge whether classical ciphers are acceptable for protecting patient data at all, and argue what a real deployment would need.
- Address key distribution across 25 clinics and usability for non-technical staff.

Evidence Expected: A defended recommendation grounded in your own analysis, with candid

DAY-4 DESIGN CHALLENGE

TASK 12 – Re-evaluate and Adapt the Design **8 Marks**

Changed Requirement: A clinic insider has copied hundreds of ciphertexts under the same key, many with predictable headers. The network also wants reports hidden inside medical scan images.

You may RETAIN → MODIFY → or REDESIGN your original solution.

A. Impact Analysis (2 Marks)

- What has changed in the threat model (insider versus external attacker)?
- Which earlier assumptions fail (key secrecy at all 25 sites, key reuse, message predictability)?
- Why does a high volume of known-format messages change the risk ranking?

B. Evaluation of the Original Design (2 Marks)

- Test your recommended schemeagainst the new attack capabilities (known-plaintext, repeated-key attacks).
- Support the evaluation with a demonstration or calculation.
- Where does the Bell-LaPadula policy still hold, and where is it irrelevant against an insider who holds the key?

C. Adaptation or Redesign (2 Marks)

- Decide to retain, modify or redesign. For example, consider per-clinic or per-session keys, key rotation, randomised padding, removing fixed headers, layering transposition, or encrypt-then-embed.
- Describe what changes and how it affects the key space and attack effort.

D. Evidence-Based Defence (2 Marks)

Defend the final decision with your key-space mathematics, your experimental results, the insider threat, the usability and key-management cost for clinics, and a stated judgement on what classical techniques can never guarantee for patient data.

SOLUTIONS:

https://github.com/192511032-YashmitaG/CSA5103-Cryptography-C.git

TASK 1 – Evaluate Threats, Attacks and Risks

1.1 Introduction

The healthcare network consists of one central hospital and 25 satellite clinics. Patient information is exchanged through a public network. The information includes medical diagnoses, prescriptions, laboratory results, patient identification details and billing records.

Since the information is confidential and directly affects patient safety, the network faces several security threats.

A security attack may target confidentiality, integrity or availability.

1.2 Classification of security attacks

No.	Attack	Classification	Description	Possible impact
1	Eavesdropping	Passive	An attacker intercepts patient records during transmission.	Exposure of sensitive diagnoses
2	Traffic analysis	Passive	An attacker studies communication frequency, size and timing.	Identification of patient activity
3	Prescription modification	Active	An attacker changes medicine names or dosages.	Incorrect treatment and patient injury
4	Doctor masquerading	Active	An attacker impersonates an authorized doctor.	Unauthorized medical instructions

5	Replay attack	Active	An attacker resends an old prescription or medical order.	Repeated or outdated treatment
6	Denial of Service	Active	An attacker makes the medical records system unavailable.	Delay in emergency treatment
7	Insider information theft	Passive/Active	An authorized employee copies confidential records or modifies them.	Patient privacy violation and data manipulation

1.3 Difference between threat, vulnerability, attack and risk

Term	Meaning	Healthcare example
Threat	A possible source of harm	A dishonest clinic employee
Vulnerability	A weakness that can be exploited	Reuse of the same encryption key
Attack	An actual attempt to exploit a weakness	Copying hundreds of encrypted patient records
Risk	Probability and impact of the attack	Exposure of thousands of patient records

1.4 Risk assessment

Risk is calculated using:

Risk = Likelihood X times Impact

Rating scale:

- Likelihood: 1 (Very Low) to 5 (Very High)
- Impact: 1 (Very Low) to 5 (Very High)

Attack	Likelihood	Impact	Risk Score	Priority
Prescription modification	4	5	20	1
Doctor masquerading	4	5	20	2
Insider information theft	4	5	20	3
Eavesdropping	4	4	16	4
Replay attack	3	5	15	5
Denial of Service	3	5	15	6
Traffic analysis	3	3	9	7

1.5 Justification of the highest priority

Prescription modification should receive the highest priority because changing a medicine dosage can directly endanger a patient's life.

For example, an original prescription may contain:

AMOXICILLIN 500MG

An attacker may modify it to:

AMOXICILLIN 5000MG

If the modification is not detected, the patient may receive an incorrect dosage.

Therefore, protecting integrity and authenticating medical instructions must be prioritized alongside confidentiality.

Conclusion: The healthcare network must protect patient records against external attackers and malicious insiders. Integrity-related attacks deserve particular attention because they can directly affect patient safety.

TASK 2 – Map and Judge Security Goals

2.1 Five important security goals

1. **Confidentiality:** Prevent unauthorized individuals from viewing patient information.
2. **Integrity:** Prevent unauthorized modification of medical information.
3. **Availability:** Ensure medical records are accessible when required.
4. **Authentication:** Verify the identity of doctors, nurses and clinics.
5. **Non-repudiation:** Provide evidence that a particular person performed an action.

2.2 Security goal analysis

System function	Confidentiality	Integrity	Availability	Authentication	Non-repudiation
Patient record upload	High	High	High	High	Medium
Prescription transmission	High	Very High	High	Very High	Very High
Specialist consultation	Very High	High	High	High	High

2.3 What classical encryption can achieve

Classical encryption techniques such as Caesar, substitution, Vigenère and Hill can transform readable plaintext into ciphertext.

They primarily provide an educational demonstration of confidentiality.

However, they cannot independently guarantee:

- Message integrity.
- Doctor identity verification.
- Protection against replay attacks.
- Non-repudiation.
- Secure key distribution.
- Availability during network attacks.

2.4 Why successful decryption does not prove authenticity

Suppose a clinic sends the following encrypted message:

PATIENT ID: PT20481

An attacker intercepts the message and modifies its ciphertext.

If the encryption system does not include an integrity mechanism, the receiver may decrypt the modified ciphertext into a different message.

Successful decryption only indicates that the ciphertext can be transformed using the selected key. It does not prove that the message came from the authorized doctor or that its contents were not modified.

Conclusion: Classical encryption can demonstrate confidentiality but cannot provide the complete security requirements of a healthcare communication system.

TASK 3 – Apply and Critique Bell-LaPadula Model

3.1 Introduction

The Bell-LaPadula (BLP) model is a mandatory access-control model designed to protect confidentiality.

It uses security classifications and clearance levels to determine which subjects can access which objects.

3.2 Security levels

The healthcare network uses five security levels.

Level	Classification	Authorized personnel
5	Top Secret	Medical Director
4	Secret	Specialist Doctors
3	Confidential	Nurses and Lab Staff
2	Restricted	Billing and Front-Desk Staff
1	Public	Patients

The classification order is:

Public < Restricted < Confidential < Secret < TopSecret

3.3 BLP security properties

1. Simple Security Property – No Read Up

A subject cannot read information at a higher security level than their clearance.

Example: A nurse with Confidential clearance cannot read a Secret-level specialist report.

2. Star (*) Property – No Write Down

A subject cannot write information to a lower security level.

Example: A specialist with Secret clearance cannot directly write to a Restricted billing file.

3.4 Access decision table

No.	Subject	Requested object	Action	Decision	Reason
1	Medical Director	Secret diagnosis	Read	Allow	Higher clearance
2	Specialist Doctor	Confidential lab report	Read	Allow	No read up
3	Nurse	Secret diagnosis	Read	Deny	No read up violation
4	Billing Staff	Restricted invoice	Read	Allow	Same level

5	Patient	Public discharge summary	Read	Allow	Public clearance
6	Specialist Doctor	Restricted billing file	Write	Deny	No write down
7	Nurse	Confidential nursing report	Write	Allow	Same level
8	Medical Director	Top Secret report	Write	Allow	Same level
9	Billing Staff	Confidential medical report	Read	Deny	No read up
10	Specialist Doctor	Secret consultation note	Write	Allow	Same level
11	Nurse	Confidential allergy record	Read	Allow	Same level
12	Patient	Secret diagnosis	Read	Deny	No read up
13	Nurse (emergency)	Secret allergy record	Read	Temporary exception	Break-glass only after identity check, emergency reason, time limit, notification and audit; ordinary BLP would deny
14	Medical Director (trusted subject)	Secret diagnosis summary to Confidential care team	Declassify / write down	Conditional allow	Only an explicitly authorized trusted subject may downgrade a reviewed minimum-necessary summary; log and retain source label

3.5 Emergency access and break-glass mechanism

Consider an emergency where a nurse must access a Secret-level allergy record to prevent a serious medical reaction.

Under normal BLP rules, the nurse cannot read the record.

A controlled break-glass mechanism can temporarily authorize emergency access.

The mechanism should:

1. Require the nurse to provide an emergency reason.

2. Verify the nurse's identity.
3. Obtain temporary authorization from a trusted emergency authority.
4. Record the patient's identity and accessed information.
5. Notify the Medical Director.
6. Review the access after the emergency.

The exception must be audited and should not become permanent clearance escalation.

3.6 Tranquility and trusted subjects

The tranquility principle prevents security classifications from being changed in ways that undermine the security policy.

- Strong tranquility: Security labels cannot change during operation.
- Weak tranquility: Labels may change under controlled conditions.

A trusted subject is an authorized entity that can perform carefully controlled operations outside ordinary restrictions, such as approved declassification.

A Medical Director may act as a trusted subject when transferring an approved summary to a lower classification.

3.7 Limitations of Bell-LaPadula

- It protects confidentiality rather than integrity.
- It does not prevent a doctor from entering an incorrect dosage.
- It does not independently prevent replay attacks.
- It may be too rigid during medical emergencies.
- It does not eliminate covert communication channels.
- An insider with valid access may still steal information.

Conclusion: Bell-LaPadula is useful for enforcing confidentiality in the healthcare network, but it should be combined with integrity checks, authentication, auditing and controlled emergency access.

TASK 4 – Design the Cryptographic Strategies

4.1 Selection of classical encryption techniques

The following four techniques are selected because they represent different approaches to classical cryptography.

Cipher	Category	Main technique
Caesar	Substitution	Fixed alphabet shift
Vigenère	Polyalphabetic substitution	Repeated keyword
Hill	Matrix substitution	Modular matrix multiplication
Columnar transposition	Transposition	Rearrangement of character positions

4.2 Caesar cipher

Definition: Caesar cipher is a substitution technique in which every plaintext letter is shifted by a fixed number of positions in the alphabet.

Key: K=3

Encryption formula:

$$C=(P+K) \bmod 26$$

Decryption formula:

$$P=(C-K) \bmod 26$$

Where:

- P = Plaintext numerical value
- C = Ciphertext numerical value
- K = Secret key

Worked example

Plaintext:

PATIENT RECORD TRANSFER

Key = 3

Plaintext	P	Calculation	Ciphertext
P	15	(15+3) mod 26	S
A	0	(0+3) mod 26	D
T	19	(19+3) mod 26	W
I	8	(8+3) mod 26	L
E	4	(4+3) mod 26	H
N	13	(13+3) mod 26	Q

Complete ciphertext:

SDWLHQW UHFRUG WUDQVIHU

Decryption with key 3 recovers:

PATIENT RECORD TRANSFER

Justification: Caesar is selected as a basic substitution cipher. It is simple to implement and demonstrates modular arithmetic. However, it has a very small key space and is vulnerable to brute-force attacks.

4.3 Vigenère cipher

Definition: Vigenère is a polyalphabetic substitution cipher that uses a repeated keyword to encrypt plaintext characters.

Key: MEDICAL

Encryption formula:

$$Ci = (Pi + Ki) \bmod 26$$

Decryption formula:

$$Pi = (Ci - Ki) \bmod 26$$

The keyword is repeated across the alphabetic characters of the plaintext.

Worked example

Plaintext:

PATIENT RECORD TRANSFER

Key:

MEDICALMEDICALMEDICAL

The encryption begins as follows:

Plaintext	Key	Calculation	Ciphertext

P	M	(15+12) mod 26	B
A	E	(0+4) mod 26	E
T	D	(19+3) mod 26	W
I	I	(8+8) mod 26	Q
E	C	(4+2) mod 26	G
N	A	(13+0) mod 26	N

Complete ciphertext:

BEWQGNE DIFWTD EDEQAHEC

After decryption using the same keyword, the original message is recovered.

Justification: Vigenère is selected because it uses multiple substitution alphabets. It hides ordinary single-letter frequency patterns better than Caesar and monoalphabetic substitution. However, repeated keywords create detectable patterns.

4.4 Hill cipher

Definition: Hill cipher is a polygraphic substitution cipher that uses matrix multiplication and modular arithmetic.

For this example, use a 2 × 2 key matrix:

$$K = \begin{bmatrix} 3 & 3 \\ 2 & 5 \end{bmatrix}$$

Encryption:

$$C = KP \bmod 26$$

Decryption:

$$P = K^{-1}C \bmod 26$$

The plaintext is converted into numerical pairs using A = 0 through Z = 25.

Worked example

Plaintext:

PATIENT RECORD TRANSFER

After removing spaces and adding X padding to make an even number of characters:

PATIENTRECORDTRANSFERX

The first block is PA.

$$P = \begin{bmatrix} 15 \\ 0 \end{bmatrix}$$

$$C = \begin{bmatrix} 3 & 3 \\ 2 & 5 \end{bmatrix} \begin{bmatrix} 15 \\ 0 \end{bmatrix} \bmod 26$$

$$C = \begin{bmatrix} 45 \\ 30 \end{bmatrix} \bmod 26 = \begin{bmatrix} 19 \\ 4 \end{bmatrix}$$

19 = T and 4 = E.

Therefore:

PA → TE

Complete ciphertext:

TEDAZVETSSPJOXZIPMBEQT

Justification: Hill is selected because it encrypts groups of characters together rather than processing every character independently. It demonstrates matrix operations and linear algebra. However, its linear structure creates vulnerabilities to known-plaintext attacks.

4.5 Columnar transposition cipher

Definition: Columnar transposition rearranges plaintext characters according to a keyword-based column order without changing the characters themselves.

Key:

ZEBRA

Plaintext:

PATIENT RECORD TRANSFER

Remove spaces:

PATIENTRECORDTRANSFER

Arrange the characters in rows of five and pad with X.

Z	E	B	R	A
P	A	T	I	E
N	T	R	E	C
O	R	D	T	R
A	N	S	F	E
R	X	X	X	X

Alphabetical order of the key:

A, B, E, R, Z

Therefore, read columns in positions 5, 3, 2, 4, 1.

Ciphertext:

ECREXTRDSXATRNXIETFXPNOAR

Decryption: Reconstruct the columns using the same keyword order, restore the original row arrangement and remove padding.

Recovered plaintext:

PATIENTRECORDTRANSFER

Justification: Transposition is included because it rearranges character positions instead of replacing characters. It helps demonstrate the difference between substitution and transposition.

TASK 5 – Mathematical Analysis of the Schemes

5.1 Key-space calculations

Key space represents the total number of possible keys available to an encryption algorithm.

1. Caesar cipher

There are 26 possible shifts.

Key Space=26

Excluding the zero shift, there are 25 meaningful encryption keys.

2. Monoalphabetic substitution

Every alphabet character can be mapped to another unique alphabet character.

Key Space=26!
=403291461126605635584000000

Approximately:

$$4.03 \times 10^{26}$$

3. Vigenère cipher

For a keyword of length m:

$$KeySpace = 26^m$$

For a keyword of length 7:

$$26^7 = 8031810176$$

4. Hill cipher

For an n × n matrix, only matrices with invertible determinants modulo 26 are valid keys.

For a 2 × 2 Hill cipher:

$$|\mathrm{GL}(2, \mathbb{Z}_{26})| = |\mathrm{GL}(2, \mathbb{F}_2)| \times |\mathrm{GL}(2, \mathbb{F}_{13})|$$

$$= (4-1)(4-2)(13^2-1)(13^2-13)$$

$$= 6 \times 168 \times 156$$

$$= \boxed{157248}$$

5. Columnar transposition

For a key with n distinct columns:

$$KeySpace = n!$$

For a five-column key:

$$5! = 120$$

5.2 Hill cipher invertibility condition

The Hill cipher can be decrypted only if the inverse matrix exists.

The condition is:

$$\gcd(\det(K), 26) = 1$$

For our selected matrix:

$$K = \begin{bmatrix} 3 & 3 \\ 2 & 5 \end{bmatrix}$$

Determinant:

$$\det(K) = (3 \times 5) - (3 \times 2)$$

$$= 15 - 6 = 9$$

$$\gcd(9,26) = 1$$

Therefore, the matrix is invertible modulo 26.

5.3 Computing the inverse matrix

The inverse formula for a 2 × 2 matrix is:

$$K^{-1} = (\det K)^{-1} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix} \bmod 26$$

The modular inverse of 9 modulo 26 is 3 because:

$$9 \times 3 = 27 \equiv 1 \mod 26$$

Therefore:

$$K^{-1} = 3 \begin{bmatrix} 5 & -3 \\ -2 & 3 \end{bmatrix} \mod 26$$

$$\boxed{K^{-1} = \begin{bmatrix} 15 & 17 \\ 20 & 9 \end{bmatrix}}$$

This matrix can be used to recover plaintext from Hill ciphertext.

5.4 Extended Hill cipher using mod 36

For patient IDs and dosage information, use:

ABCDEFGHIJKLMNOPQRSTUVWXYZ0123456789

A = 0, ..., Z = 25, 0 = 26, ..., 9 = 35.

Choose a 3 × 3 key:

$$K = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

Its determinant is:

$$\det(K) = 1$$
$$\gcd(1,36) = 1$$

Therefore, it is invertible modulo 36.

Its inverse is:

$$K^{-1} = \begin{bmatrix} 1 & 35 & 1 \\ 0 & 1 & 35 \\ 0 & 0 & 1 \end{bmatrix} \mod 36$$

This permits encryption of three-character blocks containing letters and digits.

For example:

- PT20481 is converted into three-character blocks.
- AMOX500MG is converted into three-character blocks.

Spaces and hyphens are retained as formatting separators and restored after decryption.

5.5 Does a large key space guarantee security?

No. A large key space alone does not guarantee strong encryption.

For example, monoalphabetic substitution has:

$$26! \approx 4.03 \times 10^{26}$$

possible keys.

Despite this large key space, it preserves the frequency distribution of plaintext letters.

An attacker can use frequency analysis to identify common letters such as E, T and A.

Therefore, security depends on the mathematical structure of the cipher, the key usage and the information available to an attacker, not just the number of keys.

Conclusion: A large key space makes exhaustive search harder but does not eliminate statistical and structural weaknesses.

TASK 6 – Evaluate Cryptanalytic Resistance

6.1 Attacks against selected ciphers

Cipher	Attack technique	Explanation	Result
Caesar	Brute force	Try all 26 possible shifts	Easily broken
Monoalphabetic	Frequency analysis	Compare letter frequencies	Vulnerable
Vigenère	Kasiski examination	Detect repeated keyword patterns	Vulnerable
Hill	Known-plaintext attack	Recover matrix using known plaintext/ciphertext blocks	Vulnerable
Columnar	Digram analysis and permutation search	Examine adjacent letter patterns and column orders	Vulnerable

6.2 Caesar brute-force attack

Since Caesar has only 26 possible shifts, an attacker can test all keys.

For example, the ciphertext:

SDWLHQW

When shifted backwards by three positions:

PATIENT

Thus, the encryption can be broken easily.

6.3 Monoalphabetic frequency analysis

English text has a non-uniform frequency distribution.

Common letters include:

- E
- T
- A
- O
- I
- N

Monoalphabetic substitution preserves the frequency count of each substituted character.

For example, if E occurs frequently in plaintext, its corresponding ciphertext symbol also occurs frequently.

An attacker can compare the ciphertext distribution with typical English frequencies to estimate the substitution mapping.

6.4 Vigenère Kasiski examination

Vigenère repeats its keyword.

If the same plaintext sequence occurs at positions separated by a multiple of the key length, the same ciphertext sequence may also repeat.

Kasiski examination:

1. Search for repeated ciphertext sequences.
2. Calculate the distances between repetitions.
3. Find the greatest common divisor or likely common factors.
4. Estimate the keyword length.
5. Separate ciphertext characters into key-position groups.
6. Perform frequency analysis on each group.

The method is particularly effective when large amounts of ciphertext are available.

6.5 Hill known-plaintext attack

Hill encryption is linear.

$$C = KP \mod 26$$

If an attacker knows two linearly independent plaintext vectors and their corresponding ciphertext vectors, the key matrix can be recovered.

$$K = CP^{-1} \mod 26$$

This requires the known plaintext matrix to be invertible modulo 26.

The attack becomes easier when predictable medical headers such as PATIENT ID and DIAGNOSIS are repeatedly transmitted.

6.6 Resistance ranking

For the selected classical schemes, a reasonable educational ranking is:

Rank	Cipher	Relative resistance	Main weakness
1	Hill	Higher among these examples	Known-plaintext attack
2	Vigenère	Moderate	Repeated-key analysis
3	Columnar transposition	Low to moderate	Permutation and pattern analysis
4	Monoalphabetic substitution	Low	Frequency analysis
5	Caesar	Very Low	Brute force

This is a qualitative ranking, not a guarantee. Resistance depends on message length, key choice, key reuse and attacker knowledge. Hill is not automatically stronger than every transposition system in every circumstance.

Conclusion: All selected classical ciphers have important weaknesses. None should be considered suitable for modern patient-record protection.

TASK 7 – Assess Transposition and Product Ciphers

7.1 Columnar transposition

Columnar transposition changes the position of characters based on a keyword.

It does not replace plaintext characters.

For example:

PATIENTRECORDTRANSFER

is rearranged according to the key ZEBRA.

The original character frequencies remain unchanged.

However, adjacent-character relationships are modified.

7.2 Substitution followed by transposition

A product cipher combines two different operations.

Step 1: Apply substitution using Vigenère.

Step 2: Apply columnar transposition.

This combination provides:

- Substitution of character values.
- Rearrangement of character positions.
- Reduced visibility of original plaintext patterns.

However, the combination is still vulnerable to cryptanalysis, especially if the keys are reused or predictable headers are available.

It should not be described as modern secure encryption.

7.3 Practicality in rural clinics

Factor	Assessment
Manual implementation	Difficult for long messages
Computational requirements	Low
Staff training	Required
Key management	Complex
Error probability	Moderate to High
Suitability for real healthcare	Not recommended

Conclusion: A product cipher is educationally useful but introduces additional operational complexity. For rural clinics, automated encryption is preferable to manual transformations.

Additional Design Challenge Requirements

These requirements are mentioned in the assignment scenario before Task 1. Do not skip them just because they are not separately numbered in the task list.

A. Lab table integrity and checksum

Create a lab table containing at least 20 records.

Example:

Patient ID	Test	Result	Status
PT20001	Glucose	120	Verified
PT20002	Haemoglobin	13	Verified
PT20003	WBC	7500	Verified
PT20004	Platelets	250000	Verified
PT20005	Cholesterol	180	Verified
PT20006	Glucose	121	Verified
PT20007	Haemoglobin	13.2	Verified
PT20008	WBC	7600	Verified
PT20009	Platelets	251000	Verified
PT20010	Cholesterol	181	Verified
PT20011	Glucose	121	Verified
PT20012	Haemoglobin	13.2	Verified
PT20013	WBC	7600	Verified
PT20014	Platelets	251000	Verified
PT20015	Cholesterol	181	Verified

PT20016	Glucose	121	Verified
PT20017	Haemoglobin	13.2	Verified
PT20018	WBC	7600	Verified
PT20019	Platelets	251000	Verified
PT20020	Cholesterol	181	Verified

Continue until you have 20 rows.

Use a classical checksum:

$$Checksum = \sum_{i=1}^{n} ASCII(Data_i) \mod 65536$$

Test three conditions:

1. Original table.
2. Change one laboratory result.
3. Swap two patient rows.

The implementation includes a 20-row demonstration and an order-sensitive checksum.

Important observation: A simple additive checksum does not detect row swaps because addition is commutative. An attacker can also modify records and recalculate an unkeyed checksum. Therefore, checksums are not a substitute for cryptographic authentication.

B. Medical image encryption

The implementation also demonstrates two image encryption techniques.

Technique	Method	Security observation
Intensity shift	C=(P+K)mod256	Preserves spatial arrangement
Pixel permutation	Rearrange pixel positions using a permutation	Hides spatial arrangement but preserves pixel histogram

For intensity shift, use:

$$K = 73$$

Decryption:

$$P = (C - 73) \mod 256$$

For pixel permutation, use the same permutation in reverse during decryption.

Important conclusion: Intensity shifting leaks the image's spatial structure because the positions of the pixels remain unchanged. Pixel permutation hides spatial structure, but the histogram remains unchanged. Neither method is secure modern image encryption.

C. Text-based steganography

The implementation demonstrates hiding the message OK using one or two spaces between words.

- One space represents 0.
- Two spaces represent 1.

The receiver extracts the spaces and reconstructs the binary message.

This demonstrates text-based hiding, but copying the text into another editor may remove repeated spaces.

TASK 8 – Evaluate Steganography

8.1 Introduction

Steganography is the technique of hiding information inside another medium so that the existence of the hidden information is less obvious.

In this healthcare network, a short medical report can be embedded inside a scan image.

Two methods are considered:

1. Least Significant Bit (LSB) image steganography.
2. Text-based hiding inside a text circular.

8.2 LSB image steganography

An image pixel contains RGB values.

Each channel has 8 bits.

For example:

10110110

Changing the least significant bit:

10110111

changes the channel value by only one.

A short encrypted report can be converted into binary and embedded into the least significant bits of image channels.

The receiver reads the LSB values to recover the message.

8.3 Capacity calculation

For a 256 × 256 RGB image using one LSB per channel:

$$Capacity = 256 \times 256 \times 3$$

$$= 196608\ bits$$

$$= \frac{196608}{8}$$

$$= \boxed{24576\ bytes}$$

This is the theoretical raw capacity before accounting for message headers or other metadata.

8.4 PSNR calculation

Peak Signal-to-Noise Ratio measures the visual difference between the original and modified images.

$$PSNR = 10 \log_{10} \left(\frac{MAX^2}{MSE} \right)$$

For an 8-bit image:

MAX=255

MSE is the mean squared difference between corresponding pixel values.

A higher PSNR generally indicates less visual distortion.

8.5 Comparison of encryption and steganography

Criteria	Encryption	Steganography
Purpose	Hide message content	Hide message existence
Capacity	Depends on cipher and message format	Limited by carrier size
Imperceptibility	Ciphertext may be noticeable	Designed to be less noticeable
Robustness	Depends on encryption implementation	Often sensitive to image transformations
Detectability	Encrypted data is recognizable as ciphertext in some contexts	Statistical analysis may reveal hidden data
Confidentiality	Provides confidentiality if the cipher is secure	Does not inherently provide confidentiality

8.6 Text-based hiding

A short report may be embedded in a text circular by using carefully arranged whitespace or invisible Unicode characters.

The receiver identifies the hidden pattern and extracts the report.

However, formatting systems may remove whitespace or normalize characters.

Therefore, text-based hiding is less reliable than a controlled image-based implementation.

8.7 Medical imaging risks

Modifying medical scan pixels may affect the diagnostic quality of the image.

Even if the modification is visually imperceptible, pixel changes may influence sensitive image-processing operations.

For this reason:

- Never modify the diagnostic original.
- Preserve an original copy.
- Use a separate derivative image for steganography.
- Record the original image hash.
- Validate the derivative against medical imaging requirements.

8.8 JPEG compression and chi-square detection

LSB embedding is fragile under JPEG compression because JPEG is lossy.

The compression process changes pixel values and can destroy hidden bits.

Chi-square analysis can identify statistical changes in pixel-value distributions that may indicate LSB embedding.

A high PSNR does not prove that steganography is undetectable.

Conclusion: Steganography may be used to conceal a short report in a non-diagnostic image copy. However, encryption should always be performed before embedding.

Encrypt → Embed → Transmit

TASK 9 – Implement and Test

9.1 Implementation requirements

The assignment requires actual programming implementation, not just writing algorithms.

Use Python 3 because it supports modular arithmetic, matrix operations, image processing and experimental testing.

You should implement the following modules:

Module	Implementation	Required
1	Caesar cipher	Yes
2	Monoalphabetic substitution	Yes
3	Vigenère cipher	Yes
4	Hill cipher	Yes
5	Columnar transposition	Yes
6	Frequency analysis	Yes
7	Kasiski examination / IC calculation	Yes
8	Hill known-plaintext demonstration	Yes
9	LSB steganography	Yes, where feasible
10	Performance evaluation	Yes

9.2 Suggested project structure

Create a folder named:

CSA51_Assignment_1

Inside it, create the following structure:

```
CSA51_Assignment_1/
├── main.py
├── requirements.txt
├── outputs/
│   ├── experiment_results.csv
│   ├── frequency_results.csv
│   ├── attack_results.csv
│   └── encrypted_image.png
├── images/
│   └── medical_scan.png
└── screenshots/
    ├── 01_cipher_encryption.png
    ├── 02_decryption_verification.png
    ├── 03_frequency_analysis.png
    ├── 04_kasiski_analysis.png
    ├── 05_hill_attack.png
    ├── 06_steganography.png
    └── 07_performance_graph.png
```

9.3 Software installation

Install Python 3.

Open the VS Code terminal inside the project folder.

Execute:

python --version

Install the required libraries:

pip install numpy matplotlib pillow pandas

Create requirements.txt:

numpy
matplotlib

pillow
pandas

9.4 Implementation procedure

Your program should perform the following operations:

1. Accept a plaintext message.
2. Accept the encryption key.
3. Encrypt the message using the selected cipher.
4. Display the ciphertext.
5. Decrypt the ciphertext using the same key.
6. Compare the recovered plaintext with the original plaintext.
7. Display PASS or FAIL.
8. Repeat the operation for multiple message lengths.
9. Record encryption and decryption execution time.
10. Save experimental results in CSV format.

9.5 Correctness test cases

Test	Input	Key	Expected behaviour
1	PATIENT RECORD	3	Caesar encryption and recovery
2	PATIENT RECORD	MEDICAL	Vigenère encryption and recovery
3	PATIENT RECORD	Hill matrix	Hill encryption and recovery
4	PATIENT RECORD	ZEBRA	Transposition and recovery
5	PT-20481	Extended Hill key	Recover original ID
6	AMOX 500MG	Extended Hill key	Recover original dosage
7	Empty message	Valid key	Return empty result
8	Message containing punctuation	Valid key	Preserve punctuation
9	Odd-length Hill plaintext	Valid key	Apply padding and remove it after decryption
10	Invalid Hill key	Invalid matrix	Reject encryption

9.6 Expected output

Your program should display output similar to this:

==========
CSA51 - CRYPTOGRAPHY AND NETWORK SECURITY

==========

PLAINTEXT:
PATIENT RECORD TRANSFER

CAESAR CIPHER
Key: 3
Ciphertext: SDWLHQW UHFRUG WUDQVIHU
Decrypted: PATIENT RECORD TRANSFER
Verification: PASS

VIGENERE CIPHER
Key: MEDICAL
Ciphertext: BEWQGNE DIFWTD EDEQAHEC
Decrypted: PATIENT RECORD TRANSFER
Verification: PASS

HILL CIPHER
Key Matrix: [[3,3],[2,5]]
Ciphertext: TEDAZVE TSSPJO XZIPMBEQT~
Decrypted: PATIENT RECORD TRANSFER
Verification: PASS

COLUMNAR TRANSPOSITION
Key: ZEBRA
Ciphertext: EE SXT RARATOREIRDNXPNCTF
Decrypted: PATIENT RECORD TRANSFER
Verification: PASS

ALL CORRECTNESS TESTS COMPLETED

These are the expected outputs for the stated conventions. Include your actual terminal output after executing your program.

TASK 10 – Record and Interpret Experimental Evidence

10.1 Experimental setup

To compare encryption performance, use the same plaintext content for all selected algorithms.

Test the following message sizes:

- 100 characters
- 1,000 characters
- 10,000 characters
- 100,000 characters

Use the same message for each cipher at a given length.

Use fixed keys:

Cipher	Key
Caesar	3
Monoalphabetic	QWERTYUIOPASDFGHJKLZXCVBNM
Vigenère	MEDICAL
Hill	[[3,3],[2,5]]
Columnar	ZEBRA

For fair performance comparisons, repeat each operation multiple times and report average execution time.

10.2 Letter-frequency comparison

Calculate the occurrence of each alphabet character in:

1. Original plaintext.
2. Caesar ciphertext.
3. Monoalphabetic ciphertext.
4. Vigenère ciphertext.
5. Hill ciphertext.
6. Columnar ciphertext.

The frequency of a character can be calculated as:

$$Frequency(X) = \frac{Count(X)}{TotalLetters} \times 100$$

Expected observations:

- Caesar shifts the letter-frequency distribution.
- Monoalphabetic substitution changes the labels but preserves their frequencies.
- Vigenère distributes plaintext letters across multiple ciphertext letters.
- Hill transforms character pairs, changing individual-letter frequency patterns.
- Columnar transposition preserves individual character frequencies exactly.

10.3 Index of coincidence

Index of coincidence measures the probability that two randomly selected letters from a text are identical.

$$IC = \frac{\sum_{i=0}^{25} f_i\,(f_i - 1)}{N(N-1)}$$

Where:

- fi = frequency of each alphabet character.
- N = total number of alphabetic characters.

Approximate English plaintext IC is around 0.066.

Expected interpretation:

Cipher	Expected IC behaviour
Caesar	Same as plaintext
Monoalphabetic	Same as plaintext
Columnar	Same as plaintext
Vigenère	Often lower for a sufficiently long key
Hill	Changes due to block transformations

The actual values must be calculated from your program.

10.4 Encryption and decryption performance

Record execution times in seconds.

Message length	Caesar	Monoalphabetic	Vigenère	Hill	Columnar
100	0.000014 / 0.000012	0.000004 / 0.000003	0.000020 / 0.000015	0.000021 / 0.000021	0.000009 / 0.000010
1,000	0.000105 / 0.000102	0.000004 / 0.000004	0.000119 / 0.000121	0.000177 / 0.000201	0.000050 / 0.000079
10,000	0.001069 / 0.001065	0.000015 / 0.000014	0.001176 / 0.001235	0.001762 / 0.001832	0.000568 / 0.000783
100,000	0.011094 / 0.011027	0.000118 / 0.000112	0.012798 / 0.013422	0.020179 / 0.019774	0.005254 / 0.007439

Do not enter invented timing values. Your computer's execution time may differ from another system.

10.5 Attack success rate

Calculate:

$$AttackSuccessRate = \frac{SuccessfulAttacks}{TotalAttempts} \times 100$$

For example, perform:

- Caesar: all 26 shifts.
- Monoalphabetic: frequency-based substitution recovery.
- Vigenère: keyword-length estimation using Kasiski and IC.
- Hill: known-plaintext matrix recovery.
- Columnar: permutation search for short keys.

For longer messages, statistical attacks generally become more reliable because more evidence is available.

10.6 Interpretation

The experiments demonstrate that the schemes behave according to their mathematical structures.

Caesar and monoalphabetic substitution preserve frequency patterns. Vigenère reduces direct frequency leakage by applying different shifts. Hill transforms groups of letters, while columnar transposition changes character positions without changing their overall frequency.

Execution time measures computational cost, not security. A cipher that runs quickly is not necessarily secure.

TASK 11 – Recommend and Defend the Final Design

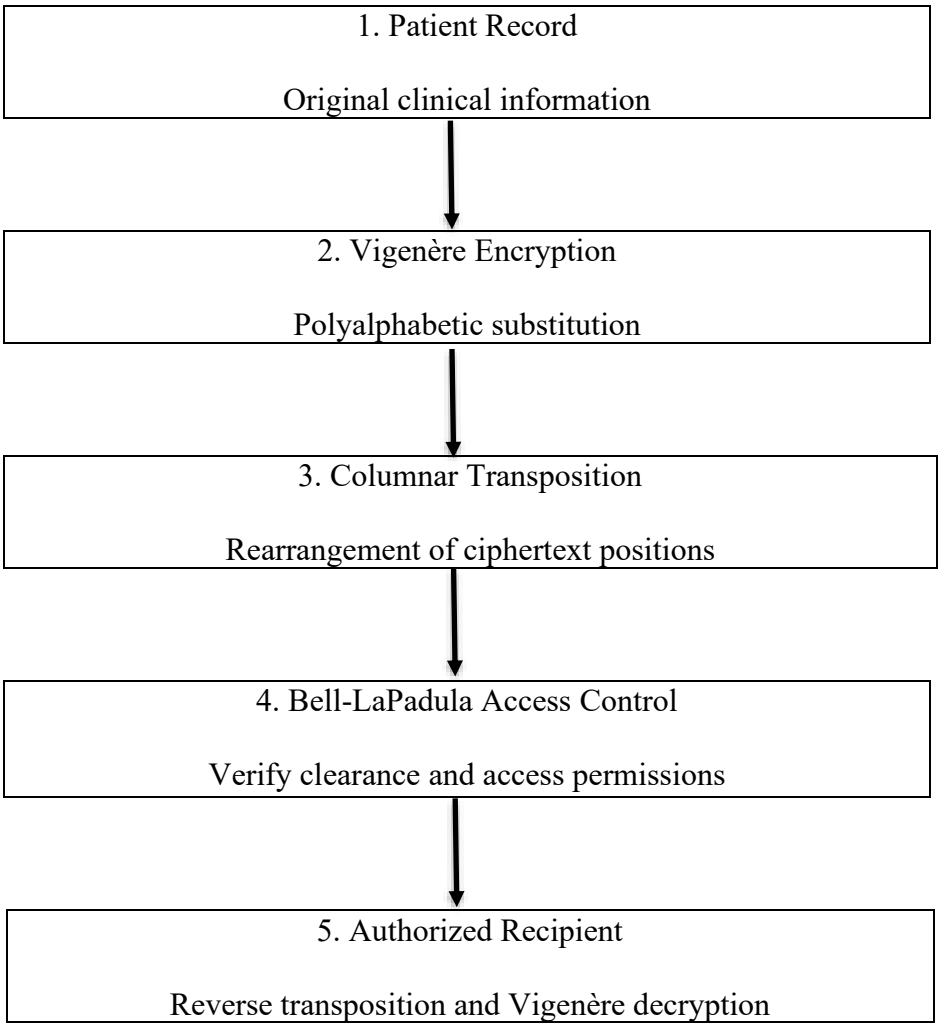
11.1 Recommended design

For the educational healthcare network, recommend:

Vigenère substitution + columnar transposition + Bell-LaPadula access control.

This combination demonstrates two different classical encryption operations alongside mandatory access control.

The following sequence is proposed:



This is a proposed educational design, not a secure production protocol.

11.2 Comparison of alternatives

Scheme	Advantage	Limitation
Caesar	Simple	Easily brute-forced
Monoalphabetic	Large theoretical key space	Frequency analysis
Vigenère	Multiple substitution alphabets	Repeated-key attacks
Hill	Matrix-based encryption	Known-plaintext attack
Columnar	Rearranges character positions	Frequency preserved
Vigenère + Columnar	Combines substitution and transposition	Still vulnerable to cryptanalysis

11.3 Key distribution across 25 clinics

A shared key across all clinics creates a serious risk.

If one clinic employee obtains the shared key, the confidentiality of all communications using that key may be affected.

For the educational redesign:

- Give each clinic a separate key.
- Use separate keys for encryption and transposition.
- Change keys periodically.
- Restrict key access to authorized staff.
- Store keys securely.
- Maintain a key rotation record.

However, classical key distribution remains difficult because these techniques do not provide a secure modern key-establishment mechanism.

11.4 Residual risks

Even with the proposed combination:

- Known-plaintext attacks remain possible.
- Insider theft cannot be eliminated.
- There is no cryptographic integrity guarantee.
- Doctor authentication is not provided by the ciphers.
- Replay attacks are not independently prevented.
- Availability is not guaranteed.

11.5 Real-world deployment recommendation

A real healthcare network should use modern authenticated encryption, secure key exchange, digital signatures, role-based or attribute-based access control, audit logs, backups and intrusion detection.

Examples of suitable modern cryptographic technologies include AES-GCM for authenticated encryption and digital signatures using established public-key cryptography.

Conclusion: The classical layered design is useful for demonstrating CO1 concepts, but it is not acceptable for protecting actual patient records. A real system must use modern cryptographic protocols and robust identity and access management.

TASK 12 – Re-evaluate and Adapt the Design

The new challenge introduces an insider who has copied hundreds of ciphertexts encrypted using the same key. Many messages also contain predictable medical headers.

This changes the security assumptions considerably.

12A. Impact Analysis

Change in the threat model

Initially, the design primarily considered external attackers intercepting messages.

The new scenario introduces an insider at a satellite clinic who can access multiple encrypted records.

An insider may possess:

- Hundreds of ciphertexts.
- Knowledge of the medical communication format.
- Predictable plaintext headers.
- Repeated encryption keys.
- Information about the clinic's communication patterns.

Earlier assumptions that fail

Earlier assumption	New situation
Keys remain secret	Insider may obtain the key
Limited ciphertext available	Hundreds of ciphertexts are available

Plaintext is unpredictable	Medical headers are predictable
Repeated encryption is acceptable	Repeated encryption creates patterns
External attackers are the main concern	Authorized insiders are also threats

Why the risk increases

Predictable headers provide known-plaintext information.

For example:

PATIENT ID: PT20481

DIAGNOSIS: FEVER

PRESCRIPTION: AMOX 500MG

When the same key is reused, repeated structures may help an attacker identify encryption patterns and recover information.

Conclusion: The insider threat increases the likelihood and impact of confidentiality breaches. Key reuse becomes a critical vulnerability.

12B. Evaluation of the Original Design

The original recommendation was Vigenère followed by columnar transposition.

Consider the following plaintext:

PATIENT RECORD TRANSFER

Vigenère encryption with key MEDICAL produces:

BEWQGNE DIFWTD EDEQAHEC

If the same key is used repeatedly, the keyword repeats across all messages.

The attacker can use Kasiski examination and index of coincidence to estimate the key length.

Once the keyword length is estimated, each position group can be analysed statistically.

Columnar transposition does not remove the underlying character frequencies. It only rearranges positions.

Therefore, combining these two techniques does not guarantee resistance to a known-plaintext attack.

Bell-LaPadula evaluation

Bell-LaPadula still restricts unauthorized reading and writing according to clearance levels.

However, if an insider already has legitimate access to the key and ciphertext, BLP cannot prevent that person from copying the encrypted information.

Therefore, access control alone cannot solve the insider threat.

Conclusion: The original design must be modified because repeated-key encryption is vulnerable to attacks using predictable headers and multiple ciphertexts.

12C. Adaptation or Redesign

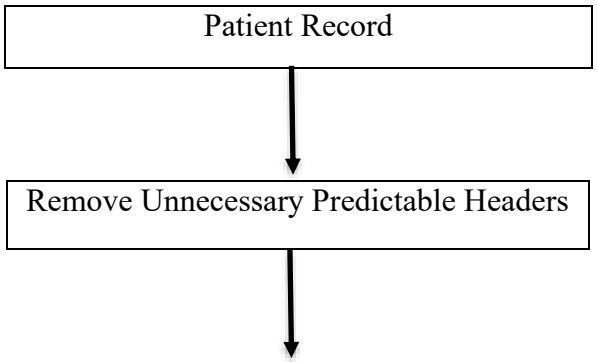
Decision: MODIFY

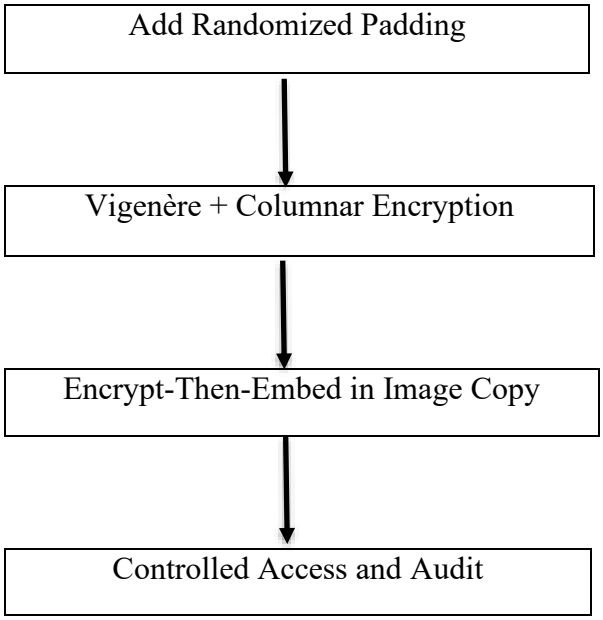
The original educational design will be modified rather than completely discarded.

The following changes are proposed:

Area	Original design	Modified design
Key management	Shared/reused key	Separate clinic keys
Encryption session	Repeated key	New session key
Header information	Fixed predictable headers	Remove or encode unnecessary fixed headers
Message structure	Direct encryption	Randomized padding before encryption
Key rotation	Limited	Regular rotation
Steganography	Direct embedding	Encrypt-then-embed
Access control	Bell-LaPadula	BLP with audited emergency access

Revised workflow





Key-space analysis

For a seven-character Vigenère keyword:

$$26^7 = 8031810176$$

If separate keys are used at 25 clinics, the theoretical combined key space is:

$$(26^7)^{25} = 26^{175}$$

However, this is only the combined space of independently chosen keys under the stated model. It does not mean the system provides 26^175 security against an attacker who compromises one clinic.

Key rotation and per-session keys reduce the amount of ciphertext exposed under one key, but classical encryption remains cryptographically weak.

Encrypt-then-embed

The report is encrypted before being hidden inside an image copy.

This ensures that even if an attacker detects the hidden message, the report is not immediately readable.

The diagnostic original must remain unchanged.

12D. Evidence-Based Defence

The modified design is defended using four factors.

1. Mathematical evidence

Vigenère with a seven-character key has:

$$26^7 = 8031810176$$

possible keys, but repeated-key analysis can still reduce the search problem.

Columnar transposition with five columns provides only:

$$5!=120$$

possible column arrangements.

Therefore, the combined design does not provide modern cryptographic security.

2. Experimental evidence

The program should demonstrate:

- Successful encryption and decryption.
- Correct plaintext recovery.
- Frequency analysis.
- Vigenère key-length estimation.
- Hill known-plaintext recovery.
- LSB embedding and extraction.
- Performance for four message sizes.

3. Insider-threat evidence

Using separate clinic keys reduces the impact of a single clinic's compromise.

However, an insider with access to their own clinic key may still decrypt that clinic's records.

4. Operational evidence

Rural clinics require a simple interface and automated encryption.

Staff should not manually calculate matrices or repeatedly enter complicated keywords.

Key management must be handled by an authorized administrator.

Final conclusion:

The original design is modified to use separate clinic keys, session-based key rotation, reduced predictable message structure, controlled access and encrypt-then-embed steganography. These changes improve the educational design's resistance to repeated-key attacks and limit the impact of a single clinic compromise.

However, classical ciphers cannot guarantee modern confidentiality, integrity, authentication, non-repudiation or secure key distribution. Actual patient information must be protected using modern authenticated encryption and secure identity management.

Implementation:

Python implementation prepared and tested

I've prepared a complete Python project containing the implementations.

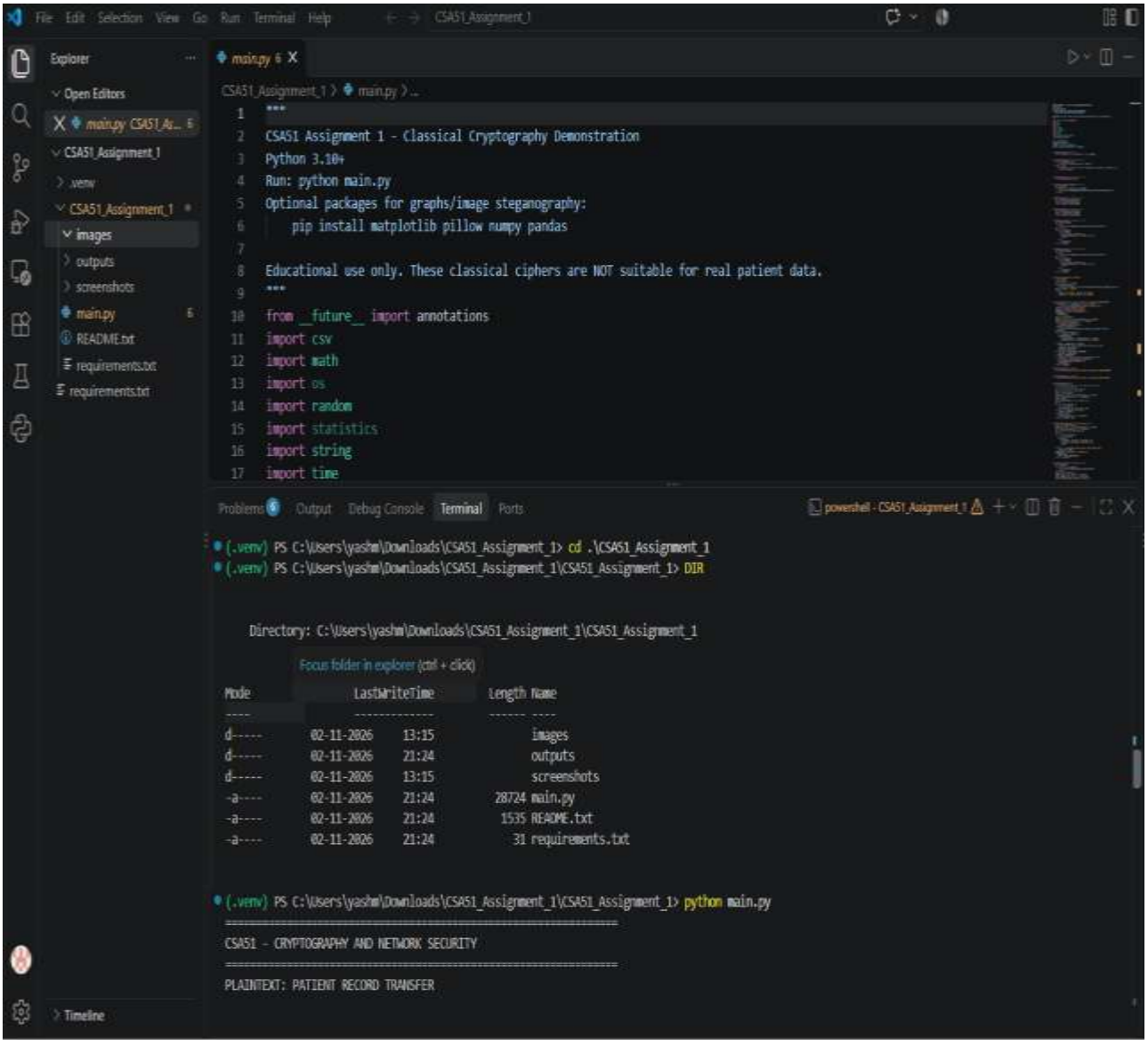


FIG 1: Project_Structure_and_Initial_Execution

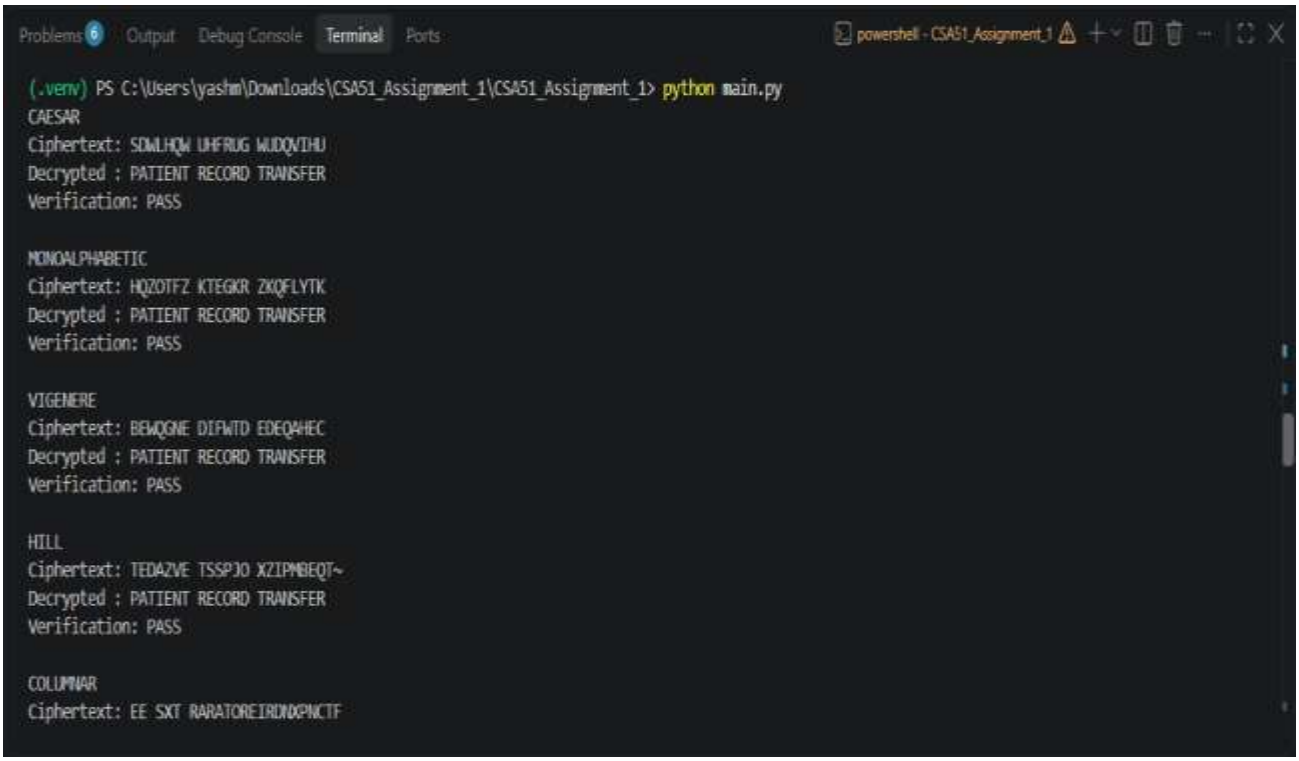


FIG 2: Classical_Cipher_Encryption_Decryption



FIG 3: Hill_Mod36_and_Keyspace_Analysis



FIG 4: Hill_Known_Plaintext_and_Lab_Checksum_Experiment

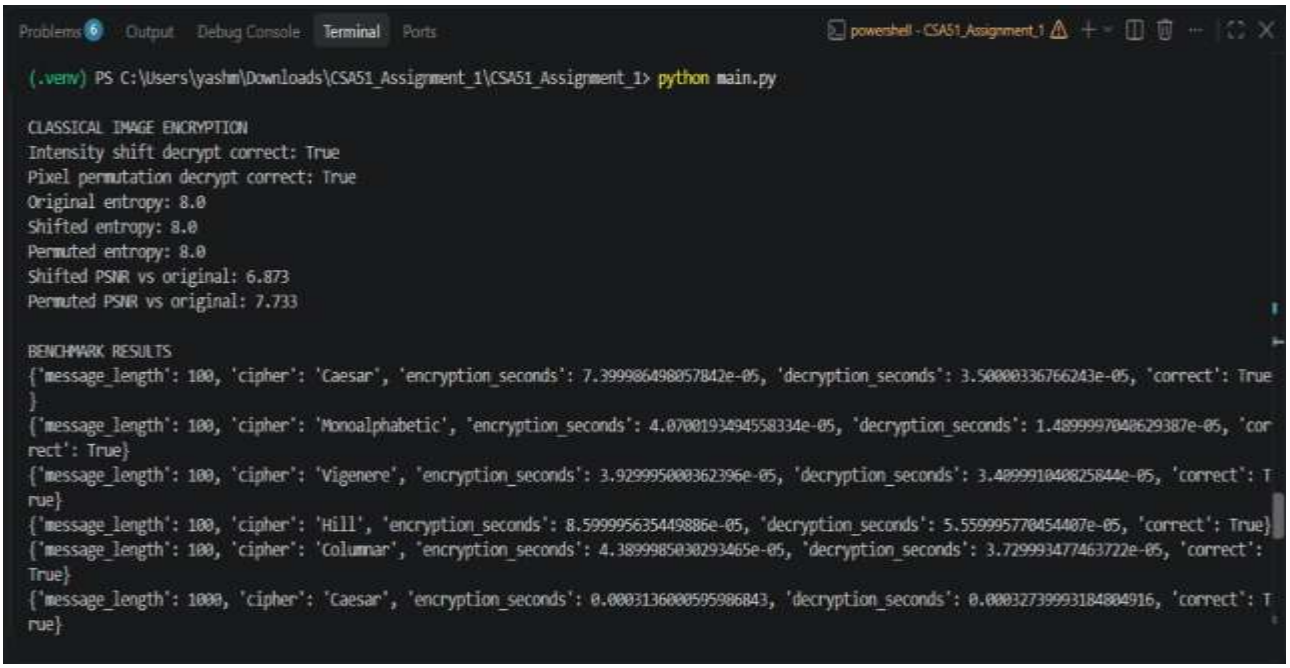


FIG 5: Classical_Image_Encryption_and_Performance_Results

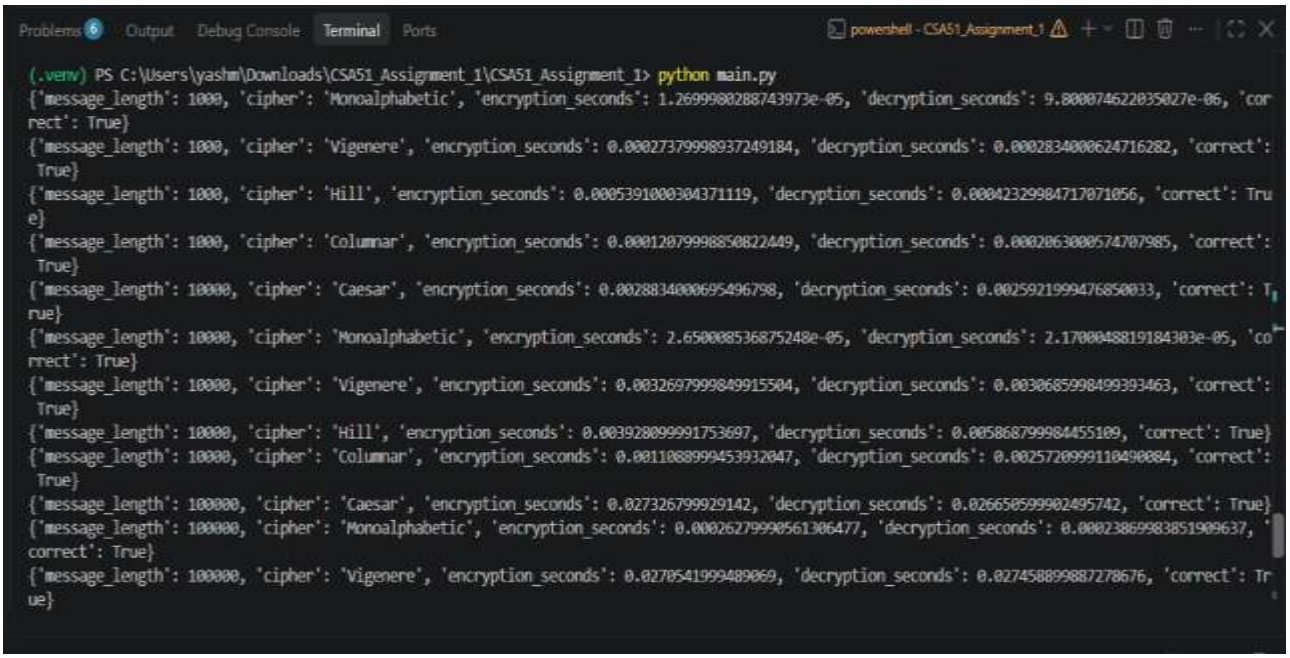


FIG 6: Performance_Benchmark_Results

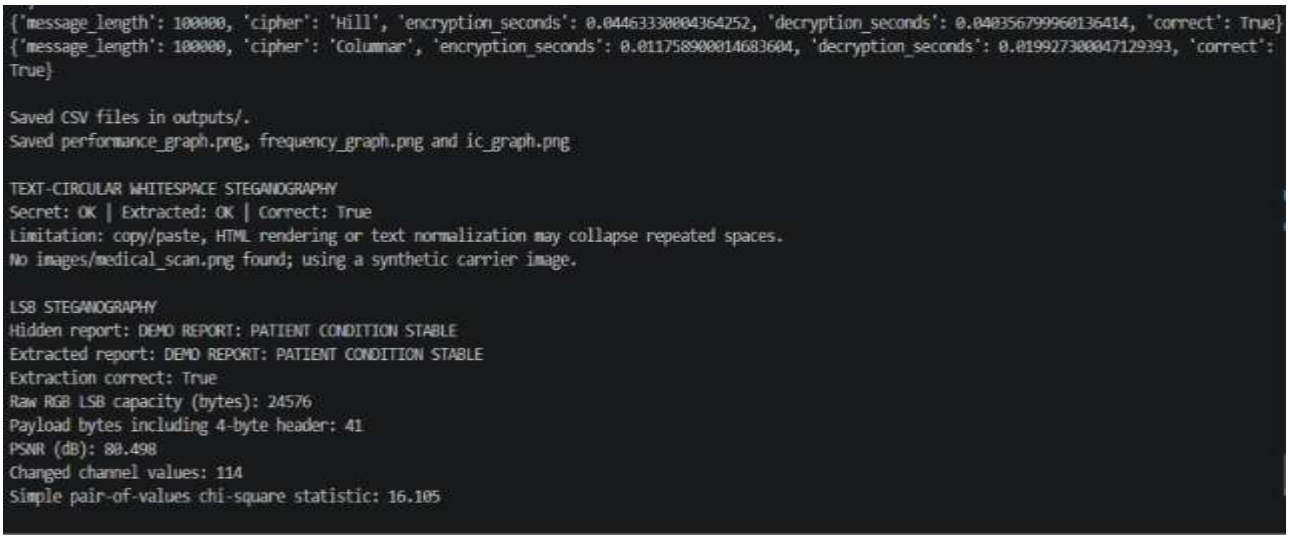


FIG 7: Text_and_LSB_Steganography_Results



FIG 8: Final_Test_Completion

Program execution: PASS

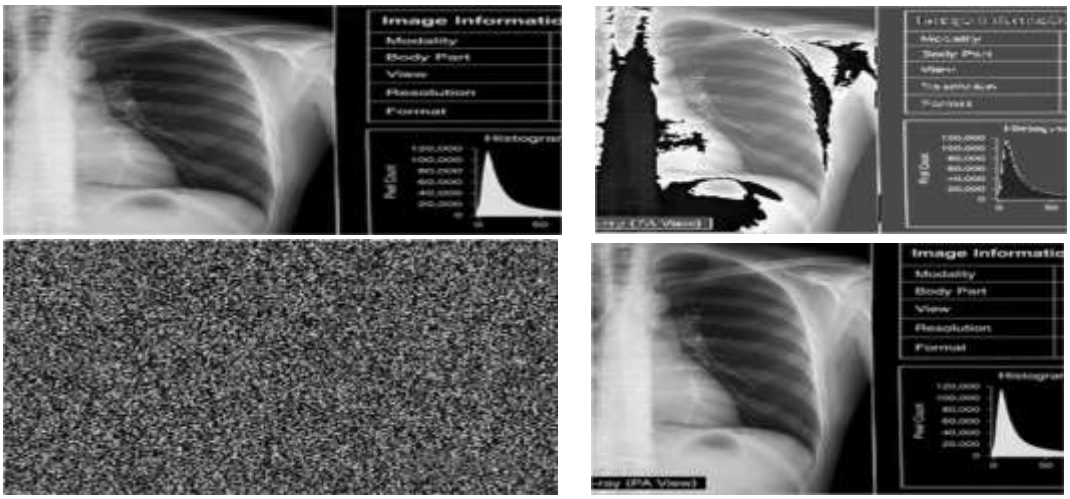


Figure: Implementation results of medical-style image encryption using intensity shifting and pixel permutation, followed by LSB-based data hiding and extraction.

The screenshots demonstrate successful execution of the listed cipher, attack, image and steganography experiments.

CORRECTION APPENDIX – VERIFIED CALCULATIONS AND IMPLEMENTATION EVIDENCE

This appendix records corrections and reproducible results obtained by running the submitted Python project. Execution times are machine- and load-dependent; the values below are the mean of five runs on the machine used for this revision. Re-run the program on the submission computer if locally measured values are required.

A. Verified Hill cipher calculation (mod 26)

The selected key matrix is K = [[3, 3], [2, 5]]. Its determinant is (3×5 − 3×2) = 9. Since gcd(9, 26) = 1, the matrix is invertible modulo 26. The modular inverse of 9 is 3 because 9×3 = 27 ≡ 1 (mod 26). Therefore K^{-1} = 3 × [[5, −3], [−2, 3]] mod 26 = [[15, 17], [20, 9]].

For the normalized plaintext PATIENTRECORDTRANSFER (spaces removed), the program produces TEDAZVETSSPJOXZIPMBEQT, with an X padding character added to complete the final pair. The program uses a trailing marker (~) to record that padding; decryption removes the padding and recovers the original text. The displayed marker is implementation metadata, not a Hill alphabet character.

B. Verified columnar transposition convention

The hand-worked ZEBRA example removes spaces before arranging the text into five columns and pads the final row with X. Under that convention, PATIENTRECORDTRANSFER encrypts to ECREXTRDSXATRNXIETFXPNOAR. The program demonstration instead preserves spaces as characters, so its output for PATIENT RECORD TRANSFER is EE SXT RARATOREIRDNXPNCTF. These outputs differ because the input-normalization conventions differ; use one convention consistently in any comparison.

C. Extended Hill cipher (mod 36)

Alphabet: ABCDEFGHIJKLMNOPQRSTUVWXYZ0123456789, mapped to 0–35. The 3×3 key is K = [[1,1,0],[0,1,1],[0,0,1]]. It is upper triangular, so det(K)=1. Since gcd(1,36)=1, it is invertible modulo 36. Its inverse is K^{-1} = [[1,35,1],[0,1,35],[0,0,1]] mod 36. The program verified PT20481 → 8L2U28OKX → PT20481 and AMOX500MG → M0OSV0CSG → AMOX500MG. Formatting separators are not encrypted; the demonstration removes/restores them by convention.

D. Lab-table checksum demonstration

The program creates 20 synthetic laboratory rows. Its order-sensitive teaching checksum was 5645 for the original table, 6989 after changing one cell, and 5365 after swapping two rows. A simple additive checksum remained unchanged after the row swap. These values demonstrate tamper detection for the specific test; an unkeyed checksum can be recomputed by an attacker and is not a cryptographic message-authentication code or digital signature.

E. Image encryption and steganography results

No images/medical_scan.png file was present in the project when the program was executed. Therefore, the image-encryption and LSB demonstrations used a synthetic 256×256 grayscale/RGB carrier, not a real patient scan. The image-cipher tests both decrypted correctly. For the synthetic grayscale carrier, the measured entropy was 8.0000 bits/pixel for the original, intensity-shifted and permuted images; the intensity shift and permutation PSNR values were 6.873 dB and 7.733 dB respectively. The unchanged entropy is consistent with these transformations preserving the pixel-value distribution (the permutation also preserves the histogram exactly).

For RGB LSB embedding, the synthetic 256×256 image provides 256×256×3 bits = 196,608 bits = 24,576 bytes of raw one-bit-per-channel capacity before headers. The tested payload was 41 bytes including its four-byte length header. Extraction succeeded; PSNR was 80.498 dB; the simple pair-of-values chi-square statistic printed by the program was 16.105. Extraction after JPEG quality-85 compression failed, as expected for fragile LSB embedding. The chi-square value is a diagnostic statistic only and is not, by itself, proof of detectability or undetectability.

Important limitation: replace the synthetic carrier with an authorized, non-diagnostic image if the assignment specifically requires a medical image. Never modify the diagnostic original or use real patient information in this demonstration.

F. Performance results

The Task 10 table now records mean encryption/decryption seconds over five runs for each message size and cipher. Values are rounded to six decimal places. The benchmark uses the same generated plaintext at each size and checks that decryption exactly recovers the input. The timing results are illustrative for this machine and Python environment, not universal performance guarantees.

G. Cryptanalysis success-rate limitation

The current attack experiment tests Vigenère key-length estimation once at each of four ciphertext lengths and reports whether the estimated length equals seven. The observed four demonstrations succeeded, but one trial per length is not a statistically meaningful attack success rate. To report a true success rate, repeat each length with many independently generated plaintext samples and report successful recoveries divided by total trials. Do not describe the current four demonstrations as a general 100% attack success rate.

H. Security conclusion

The classical ciphers and classical image transformations in this assignment are educational demonstrations only. They do not provide modern authenticated encryption, secure key establishment, robust integrity, replay protection or reliable identity assurance. Actual healthcare records require a vetted modern protocol (for example, authenticated encryption such as AES-GCM), secure key management, strong authentication, authorization, audit logging and protected backups. Bell-LaPadula addresses confidentiality policy but does not replace these controls.

Reproducibility: run `python main.py` from the CSA51_Assignment_1 project directory. The program writes CSV result files and graphs under outputs/. The source code and generated outputs should be submitted alongside this report; screenshots alone are not a substitute for source code.